

Kicked off a Horse



Section I: Scenario Demographics

Scenario Title:	Kicked off a Horse – Pelvic Fracture and SDH		
Date of Development:	20/06/2015 (DD/MM/YYYY)		
Target Learning Group:	☒ Juniors (PGY 1 – 2)	☐ Seniors (PGY ≥ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Kyla Caners
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To enhance resuscitation and team management skills through a trauma scenario with multiple key steps.
CRM Objectives:	1) Effectively lead a trauma team and delegate roles appropriately. 2) Communicate clearly as a team in a complex scenario
Medical Objectives:	1) Perform safe and thorough primary survey. 2) Demonstrate safe approach to intubating a head-injured trauma patient. 3) Recognize and manage unstable pelvic fracture and FF in abdomen a. Administer PRBCs. b. Administer TXA. c. Apply pelvic binder. d. Consult orthopedics and general surgery for definitive management.

Case Summary: Brief Summary of Case Progression and Major Events

A 32-year-old female presents after being bucked off of her horse. She is brought in as a trauma team activation because of a low BP. Her primary survey will reveal a boggy hematoma over her right temporal area as well as an unstable pelvis. Her initial GCS will be 8. The team will proceed through airway management in a hypotensive, head-injured trauma patient while also binding her pelvis.

References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.



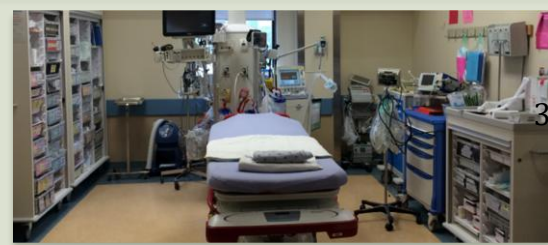
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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Confederates		Brief Description of Role	
None.			
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☒ Blood Products	☒ ET Tubes	☒ Other: Blanket to bind pelvis	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
Boggy hematoma to right temporal area. Dirty, torn clothes. Boarded and collared on arrival.			
E. Approximate Timing			
Set-Up:	10 min	Scenario:	12 min
		Debriefing:	20 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

A 32-year-old female presents as a trauma activation with EMS after being bucked off of her horse. Her mom witnessed the episode and called EMS because she seemed groggy. She has had a low BP with EMS on route. Her current BP is 80/40.

B. Patient Profile and History

Patient Name: Jennifer Starr Age: 32 Weight: 65kg

Gender: ☐ M ☒ F Code Status: Full

Chief Complaint: Bucked off a horse

History of Presenting Illness: Bucked off her horse today. EMS called when patient seems groggy.

Past Medical History:	Nil.	Medications:	OCP.

Allergies: None.

Social History: Horse trainer. Lives on a farm outside the city.

Family History: N/A

Review of Systems:	CNS:	No voiced complaints (unable).		
	HEENT:	No voiced complaints (unable).		
	CVS:	No voiced complaints (unable).		
	RESP:	No voiced complaints (unable).		
	GI:	No voiced complaints (unable).		
	GU:	No voiced complaints (unable).		
	MSK:	No voiced complaints (unable).	INT:	No voiced complaints.

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display ☒ Monitor On, no data displayed ☐ Monitor on Standard Display

HR: 120/min BP: 80/40 RR: 22/min O₂SAT: 96% RA

Rhythm: sinus tach T: 36.1°C Glucose: 7.1 mmol/L **GCS: 8 (E2 V2 M4)**

General Status: GCS 8. Appears unwell.

CNS:	As above. PERLA 3mm.		
HEENT:	Boggy hematoma over right temporal area. No other signs HI.		
CVS:	No murmur. Normal heart sounds.		
RESP:	GAEB. No adventitious. Trachea midline.		
ABDO:	Soft, NT.		
GU:	Perineal hematoma.		
MSK:	Pelvis unstable.	SKIN:	Abrasions only.

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: sinus tach HR: 120/min BP: 80/40 RR: 22/min O ₂ SAT: 96% T: 36.1°C	GCS 8. Moaning only.	<u>Learner Actions</u> - ☐ Monitors, 2 large IVs - ☐ Trauma labs (include βHCG), cap sugar (7.1) - ☐ Bolus 1L NS - ☐ Call for blood - ☐ Primary survey - ☐ CXR/Pelvic xray - ☐ FAST to r/o FF (+RUQ) - ☐ Bind pelvis - ☐ Give TXA 1g iv - ☐ Call gen sx/ortho/neurosx	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - 2L NS → BP 85/45, HR 115 <u>Triggers</u> <i>For progression to next state</i> - Intubation → 2. Intubation - 5 minutes → 3. Hypotension
2. Intubation Keep vitals same as state prior	Unchanged.	<u>Learner Actions</u> - ☐ Have vasopressor on hand - ☐ Appropriate induction (etomidate or ketamine) - ☐ Paralytic administration - ☐ Inline stabilization - ☐ ± Lidocaine/fentanyl pre-tx - ☐ Post-intubation CXR - ☐ Initiate sedation	<u>Modifiers</u> - Propofol used → BP 60/30 <u>Triggers</u> - Intubated from State 1 → 3. Hypotension - Intubated from State 3 → 4. Herniation
3. Hypotension BP → 65/35 HR → 130 If intubated → RR 12	Patient unchanged from previous state.	<u>Learner Actions</u> - ☐ Call ortho/gen sx - ☐ Bind pelvis (if not yet) - ☐ Administer 2u PRBC - ☐ Consider massive transfusion - ☐ FAST (if not yet done)	<u>Modifiers</u> - 1 unit PRBC → BP 70/35 <u>Triggers</u> - Intubation → 2. Intubation - 2 units PRBC → 4. Herniation
4. Herniation HR → 45 BP → 75/45 If not intubated → RR 6 (irregular) If intubated → RR 12	Patient's vitals reflect herniation. Left pupil blown if learners ask.	<u>Learner Actions</u> - ☐ Intubate (if not yet done) - ☐ Hyperventilate patient - ☐ HOB to 30 degrees - ☐ Hypertonic saline 100ml iv - ☐ Mannitol 500ml iv - ☐ Call neurosurgery	<u>Modifiers</u> <u>Triggers</u> - 3 interventions to reduce ICP → 5. Resolution
5. Resolution HR → 70 BP → 90/60	Unchanged	<u>Learner Actions</u> - ☐ Call neurosurgery - ☐ Facilitate conversation with neurosurgery/gen sx/ortho re: priorities for OR vs. CT vs. IR for embolization	Neurosurgery arrives → END CASE

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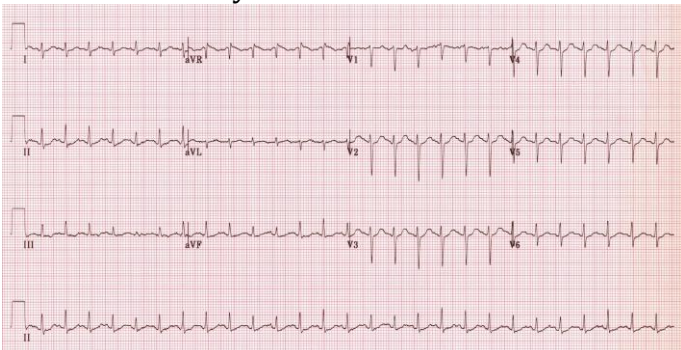
Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results

No blood work given in this case.

Images (ECGs, CXRs, etc.)

ECG – sinus tachycardia



ECG source:

<https://i0.wp.com/lifeinthefastlane.com/wp-content/uploads/2011/12/sinus-tachycardia.jpg>

CXR – pre-intubation



CXR source: <https://radiopaedia.org/cases/normal-chest-radiograph-female-1>

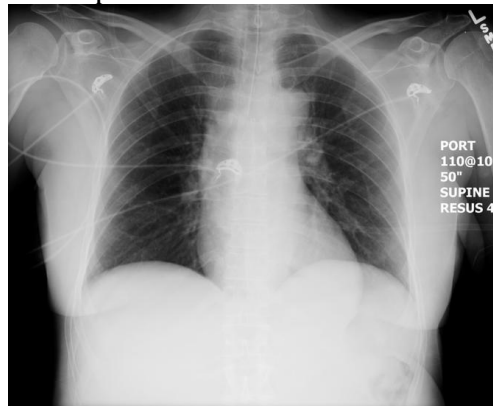
Pelvic x-ray – fracture shown



PXR source:

https://littlemedic.files.wordpress.com/2013/01/pelvis_0_1.jpg

CXR – post-intubation



CXR source:

<https://emcow.files.wordpress.com/2012/11/normal-intubation2.jpg>

Ultrasound Video Files (if applicable)

U/S showing FF RUQ

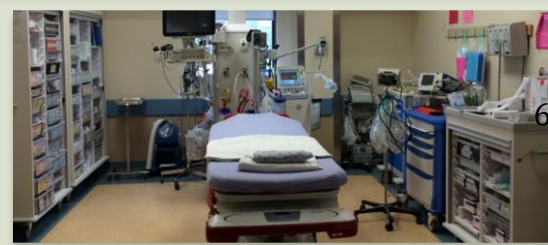
U/S showing no PCE

U/S showing normal lung sliding bilaterally



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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) How do you feel that your team handled the primary survey (initial assessment)? 2) Do the team members feel that, as a group, you responded well to a very dynamic, unstable patient? a. If not, what suggestions can you think of to improve team cohesion and management? 3) What is your approach to intubating a head-injured patient? What does “neuro-protective RSI” traditionally include? Did you consider this? What are the key priorities when intubating a head-injured patient? 4) What is permissive hypotension? Did you use it in this case? When should we not use it? 5) How did it feel having to coordinate so many priorities at once? Do you have a strategy for managing patients with multiple simultaneous priorities?	
Key Moments	
Recognition of need to intubate	
Recognition of herniation	
Management of hypotension in context of pelvic fracture and free fluid in abdomen	